

I. THE ART OF CRISIS MANAGEMENT

1.1 When, Not If

In modern Cyber Security, we operate under the "Presumed Breach" mindset. No matter how strong your firewall is, a breach will eventually occur. **Incident Response (IR)** is the organized approach to managing and addressing the aftermath of a security breach or cyberattack. At Duy Tan University, we teach students how to stay calm and methodical when the "alarm bells" go off.

1.2 The Goal of IR

- **Limit Damage:** Prevent the attack from spreading to other servers.
- **Reduce Recovery Time:** Get the business back online as quickly as possible.
- **Reduce Costs:** A well-managed incident costs significantly less than a chaotic one.

1.3 The Incident Response Life Cycle (NIST Standard)

1. **Preparation:** Creating the team and tools before an attack happens.
2. **Detection & Analysis:** Identifying that a suspicious event is actually a real attack.
3. **Containment, Eradication, & Recovery:** Stopping the attack, removing the "poison," and restoring systems.
4. **Post-Incident Activity:** Learning from the event to prevent it from happening again.

II. CAREERS IN CYBER-RESILIENCE

2.1 The Digital Firefighters

Incident responders are the elite specialists called in when a company is under active attack. This role requires high technical skill and the ability to work under extreme pressure.

2.2 Professional Roles

- **CSIRT Lead (Computer Security Incident Response Team):** The coordinator who manages the communication between technical staff, lawyers, and executives during a crisis.
- **Incident Handler:** The technical expert who "hunts" the hacker inside the network and kicks them out.
- **Disaster Recovery Coordinator:** Focuses on the "Backup" side—ensuring that if a data center is destroyed, a mirror site can take over instantly.

2.3 Industry Certifications

Students aiming for this path often pursue the GCIH (GIAC Certified Incident Handler) or the EC-Council Certified Incident Handler (ECIH).

III. BACKUP STRATEGIES & DISASTER RECOVERY (DR)

3.1 Business Continuity Planning (BCP)

While IR handles the "Attack," BCP ensures the "Business" keeps running. If a bank's main server is encrypted by ransomware, BCP is the plan that allows tellers to still process transactions using backup systems.

3.2 The 3-2-1 Backup Rule

At DTU, we drill this gold standard into every student:

- **3** copies of your data.
- **2** different media types (e.g., Cloud and Local Disk).
- **1** copy stored **Off-site** (in a different city or a secure cloud).

3.3 Recovery Site Types

- **Hot Site:** A duplicate data center with all data synced. Can take over in minutes. (Expensive).
- **Warm Site:** Has the hardware ready, but data needs to be restored from backups. Takes hours.
- **Cold Site:** Just an empty room with power and cooling. Takes days to set up.

IV. DEEP DIVE: LOG ANALYSIS & TABLETOP EXERCISES

4.1 Finding the "Patient Zero"

How did the hacker get in? Students learn to use **SIEM (Security Information and Event Management)** tools like **Splunk** or **ELK Stack** to correlate logs from firewalls, servers, and antivirus to find the starting point of the infection.

4.2 Tabletop Exercises (TTX)

Technical skills aren't enough. At DTU, we perform "Tabletop" simulations where students are given a scenario (e.g., "The University's website is defaced and student grades are being leaked"). They must decide:

- Should we shut down the server immediately?
- When do we call the police?
- How do we tell the students their data was stolen?

4.3 Practical Lab Exercise

1. **Environment:** A network of 3 Virtual Machines.
2. **Task 1: The Breach.** One machine is "infected" with a simulated worm.
3. **Task 2: Isolation.** Students must use network commands to isolate the infected VM without shutting down the healthy ones.
4. **Task 3: Evidence Preservation.** Students "clone" the hard drive of the infected machine for later forensic analysis before "cleaning" the original system.

4.4 Conclusion

A Cyber Security professional is judged not by their ability to prevent every attack, but by their ability to lead an organization through a crisis and emerge stronger. DTU graduates are the leaders who bring order to the chaos of a cyber breach.